

SIMATS ENGINEERING
ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No :192424322 Name:M.Arshini

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3)

Activity Tool : Relationship Visualization Lab

Weightage: 20%

Code of Conduct:

IM.ARSHINI(192424322) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Relationship Visualization Lab

Provided Datasets

The following files are supplied alongside this document so the lab can be completed without internet access to install extra data packages:

- `mtcars.csv` — real R built-in dataset (32 cars $\times$ 11 variables: `mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`)
- `iris.csv` — real R built-in dataset (150 flowers $\times$ 5 variables: `Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width`, `Species`)
- `gapminder2007_practice.csv` — illustrative practice dataset in the style of the Gapminder 2007 cross-section (country, continent, year, `gdpPercap`, `lifeExp`, `pop`). Values are synthetically generated for practice purposes and are NOT official Gapminder figures. If internet access is available, students may instead run `install.packages("gapminder")` and use `data(gapminder)` for the authentic dataset.
- `network_edges.csv` — custom-built edge list (15 edges, 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, Hana) with a weight column, for the network graph question.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

Question 1: Scatterplot with Regression Diagnostics

Dataset provided: `mtcars.csv`

Using the `mtcars` dataset, create a scatterplot of `mpg` vs `wt` (weight) using `ggplot2`. Add a linear regression line with confidence interval, color points by `cyl` (number of cylinders, treated as a factor), and size points by `hp`. Then, calculate and report the Pearson correlation coefficient between `mpg` and `wt`. Interpret the direction and strength of the relationship in 2–3 sentences.

Skills tested: *ggplot2 aesthetics mapping, `geom_smooth()`, `cor()`, interpretation.*

Question 2: Correlogram Construction

Dataset provided: `mtcars.csv` (all numeric columns)

Using the `mtcars` dataset, compute the correlation matrix and visualize it as a correlogram using either the `corrplot` or `GGally::ggcorr` package. Configure the plot to:

- Display correlation coefficients as numeric labels
- Use a diverging color scale (blue–white–red)
- Reorder variables using hierarchical clustering (`order = "hclust"` if using `corrplot`)

Identify the two variable pairs with the strongest positive and strongest negative correlations.

Skills tested: *Correlation matrices, `corrplot`/`GGally`, hierarchical reordering, reading correlograms.*

Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

Question 4: Bubble Chart for Multivariate Paired Data

Dataset provided: gapminder2007_practice.csv (or the gapminder package, year == 2007)

Filter for the year 2007 and create a bubble chart showing:

- x-axis: gdpPerCap (log-transformed scale)
- y-axis: lifeExp
- bubble size: pop
- bubble color: continent

Add appropriate axis labels, a title, and use `scale_size_continuous()` to control bubble size range so points don't overplot excessively. Discuss one visible cluster or outlier in the data.

Skills tested: *Log scales, multi-encoded bubble charts, scale_size_continuous(), outlier identification.*

Question 5: Hexbin Plot vs. Network Graph

This question has two parts to compare visualization strategies for different relationship types.

Part a — Hexbin Plot

Dataset provided: Simulated in R (no file needed)

Simulate two correlated variables with 10,000 observations (`set.seed(123)`, use `mnorm()` with a specified covariance structure or `MASS::mvrnorm()`). Create a hexbin plot using `geom_hex()` in `ggplot2`. Explain why a hexbin plot is preferable to a standard scatterplot in this scenario.

Part b — Network Graph

Dataset provided: network_edges.csv

Using the provided edge list (8 nodes, 15 weighted edges), build a network graph using `igraph` or `ggraph`. Size nodes by degree centrality and color edges by weight. Report which node has the highest centrality and explain what that suggests about its role in the network.

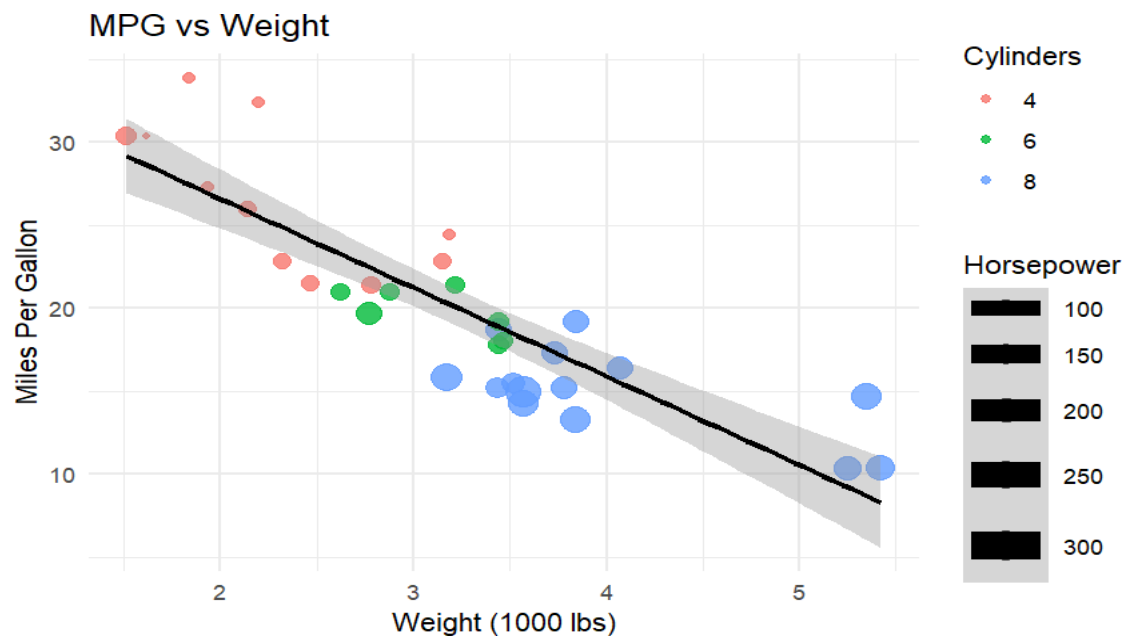
Skills tested: *geom_hex(), overplotting solutions, igraph/ggraph, centrality measures, network interpretation.*

Relationship Visualization Lab

QUESTION-1 Scatterplot with Regression Diagnostics

Aim

To visualize the relationship between miles per gallon (`mpg`) and car weight (`wt`), while also representing the number of cylinders and horsepower.



CODE:

```
library(ggplot2)
# Load built-in dataset
data(mtcars)
# Convert cyl to factor
mtcars$cyl <- as.factor(mtcars$cyl)
# Scatterplot
ggplot(mtcars, aes(x = wt, y = mpg,
                   color = cyl, size = hp)) +
```

```

geom_point(alpha = 0.8) +
geom_smooth(method = "lm", se = TRUE,
            color = "black") +
labs(
  title = "MPG vs Weight",
  x = "Weight (1000 lbs)",
  y = "Miles Per Gallon",
  color = "Cylinders",
  size = "Horsepower"
) +
theme_minimal()
# Pearson correlation
cor(mtcarsmpg, mtcarswt, method = "pearson")

```

Expected Result

The Pearson correlation coefficient is approximately:

-0.8677

Interpretation

The correlation between mpg and wt is approximately -0.868, indicating a strong negative relationship. As the weight of a car increases, its fuel efficiency in miles per gallon generally decreases. The regression line in the scatterplot also shows this downward trend.

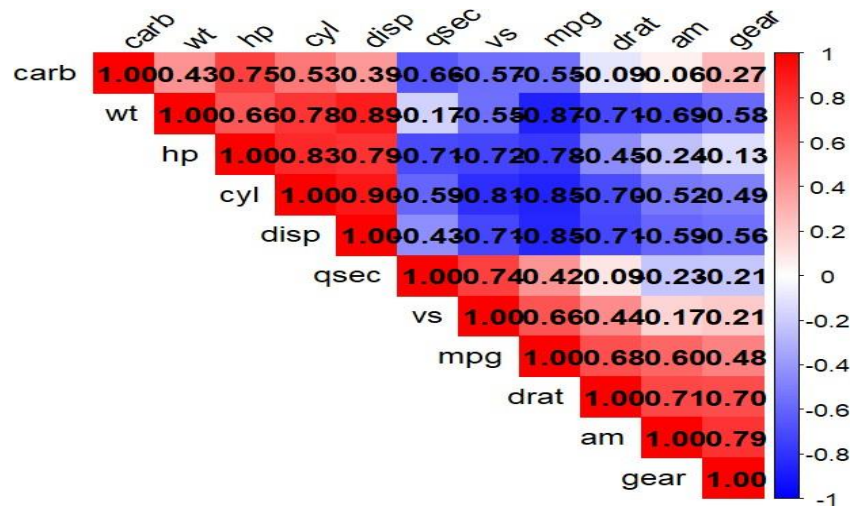
Rubric Connection

- **Relationship Visualization:** Scatterplot clearly shows `wt` vs `mpg`.
- **Relationship Analysis:** Pearson correlation measures the strength and direction.
- **Visualization Techniques:** Color represents `cyl`, size represents `hp`, and `geom_smooth()` adds regression diagnostics.
- **Clarity:** Labels, title, and legend make the graph easy to understand.

Question 2: Correlogram Construction

Aim

To calculate and visualize the correlations among all numeric variables in the `mtcars` dataset.



CODE:

```
library(corrplot)

# Load built-in dataset

data(mtcars)

# Correlation matrix

cor_matrix <- cor(mtcars)

# Display correlation matrix

print(cor_matrix)

# Correlogram

corrplot(

cor_matrix,

method = "color",

type = "upper",
```

```
order = "hclust",  
  
addCoef.col = "black",  
  
tl.col = "black",  
  
tl.srt = 45,  
  
col = colorRampPalette(  
  
  c("blue", "white", "red") )(200))
```

Expected Important Results

For the standard `mtcars` dataset:

Relationship	Variables	Approximate Correlation
Strongest Positive	<code>disp</code> and <code>cyl</code>	0.902
Strongest Negative	<code>mpg</code> and <code>wt</code>	-0.868

Depending on whether duplicated upper/lower matrix pairs are counted, the same relationship may appear twice.

Interpretation

The strongest positive relationship is generally between **engine displacement (`disp`)** and **number of cylinders (`cyl`)**, meaning cars with more cylinders tend to have larger engines. The strongest negative relationship is between **`mpg` and `wt`**, showing that heavier cars generally have lower fuel efficiency.

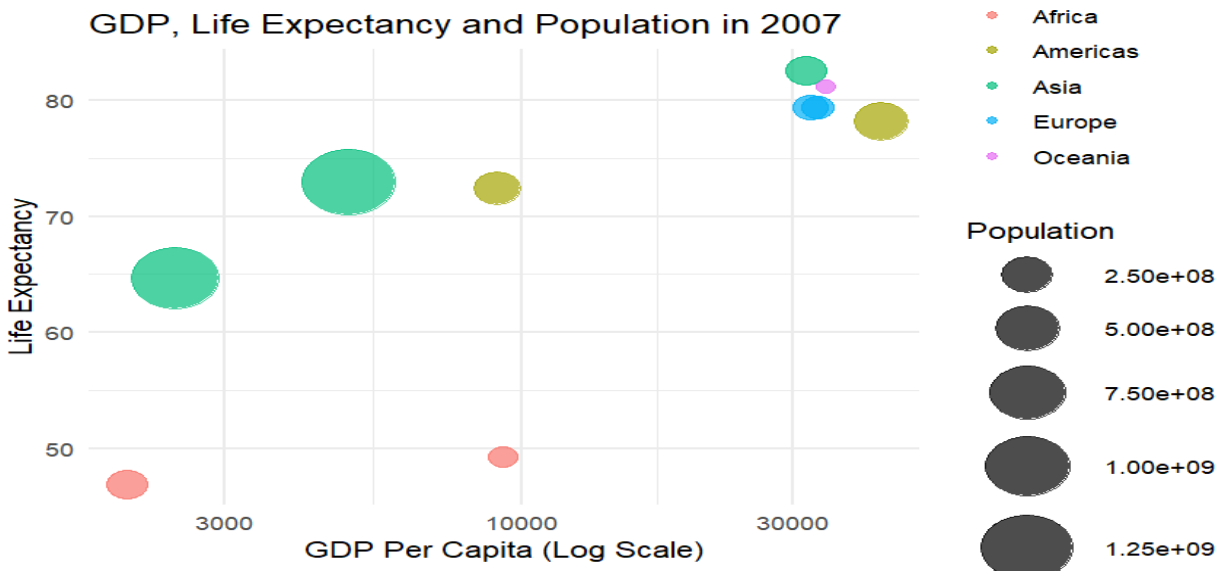
Rubric Connection

- **Relationship Visualization:** The correlogram displays relationships among all numeric variables.
- **Relationship Analysis:** Correlation coefficients quantify the strength and direction.
- **Visualization Technique:** Hierarchical clustering groups related variables together.
- **Accuracy:** Numeric labels allow exact interpretation of correlations.

Question 4: Bubble Chart for Multivariate Paired Data

Aim

To study the relationship between GDP per capita, life expectancy, population, and continent for countries in the year 2007.



CODE:

```
library(ggplot2)

# Create Gapminder-style dataset

gapminder_data <- data.frame(

  country = c(

    "India", "China", "USA", "Japan",

    "Germany", "Brazil", "Nigeria",

    "South Africa", "Australia", "UK"),

  continent = c(

    "Asia", "Asia", "Americas", "Asia",
```



```
"Europe", "Americas", "Africa",  
"Africa", "Oceania", "Europe" ),
```

```
year = c(  
  2007, 2007, 2007, 2007, 2007,  
  2007, 2007, 2007, 2007, 2007),
```

```
gdpPercap = c(  
  2452, 4959, 42952, 31656,  
  32170, 9065, 2013, 9269,  
  34435, 33203 ),
```

```
lifeExp = c(  
  64.7, 73.0, 78.2, 82.6,  
  79.4, 72.4, 46.9, 49.3,  
  81.2, 79.4),
```

```
pop = c(  
  1110396331,  
  1318683096,  
  301139947,  
  127467972,  
  82400996,  
  190010647,  
  135031164,  
  43997828,  
  20434176,  
  60776238 ))
```

```

# Filter year 2007

data_2007 <- subset(gapminder_data, year == 2007)

# Bubble chart

ggplot(
  data_2007,
  aes(
    x = gdpPercap,
    y = lifeExp,
    size = pop,
    color = continent)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  scale_size_continuous(range = c(3, 15)) +
  labs(
    title = "GDP, Life Expectancy and Population in 2007",
    x = "GDP Per Capita (Log Scale)",
    y = "Life Expectancy",
    size = "Population",
    color = "Continent" ) +
  theme_minimal()

```

Interpretation

A visible pattern is that countries with higher GDP per capita generally have higher life expectancy. Many African countries form a cluster at lower GDP and lower life expectancy, while countries in Europe and Oceania are generally concentrated at higher GDP and higher life expectancy. A major outlier is represented by a very large bubble such as **China or India**, because of their extremely large populations.

Why Log Transformation?

GDP per capita values vary greatly between countries. Using:

`scale_x_log10()`

compresses the large range of values and makes the relationship between countries easier to observe.

Rubric Connection

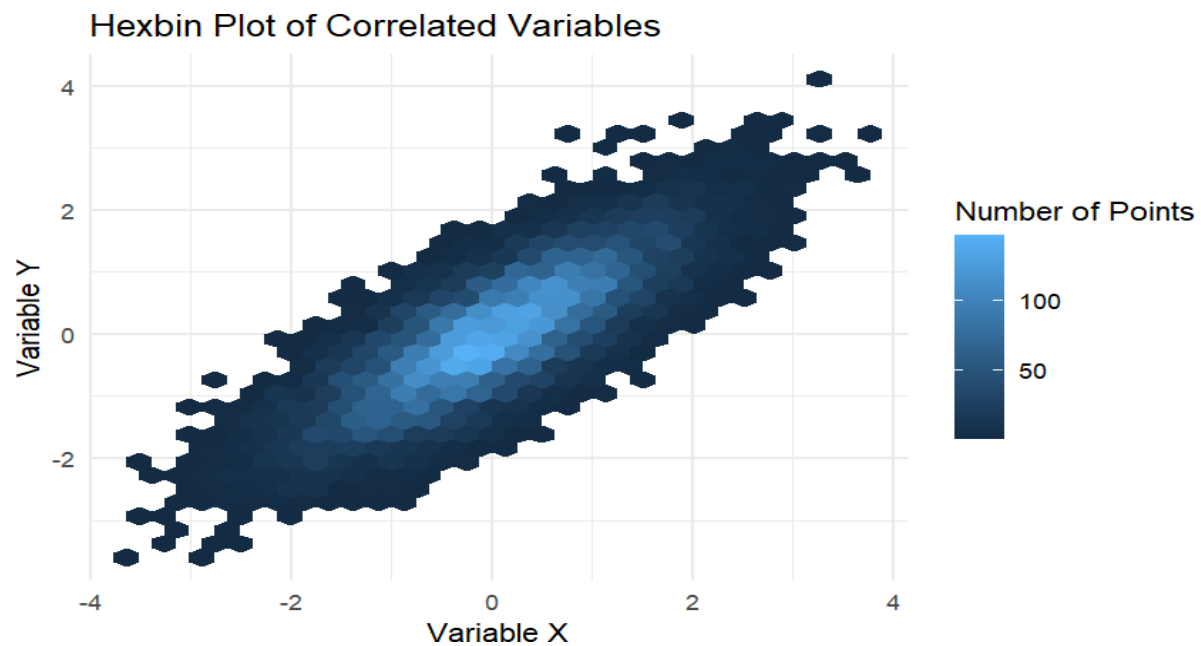
- **Relationship Visualization:** Displays four variables in one graph.
- **Relationship Analysis:** Shows the GDP–life expectancy relationship.
- **Visualization Techniques:** Uses bubble size, color encoding, and logarithmic transformation.
- **Clarity:** Axis labels and legends explain every variable.

Question 5: Hexbin Plot vs. Network Graph

Part A — Hexbin Plot

Aim

To visualize the relationship between two correlated variables containing **10,000 observations**.



Interpretation

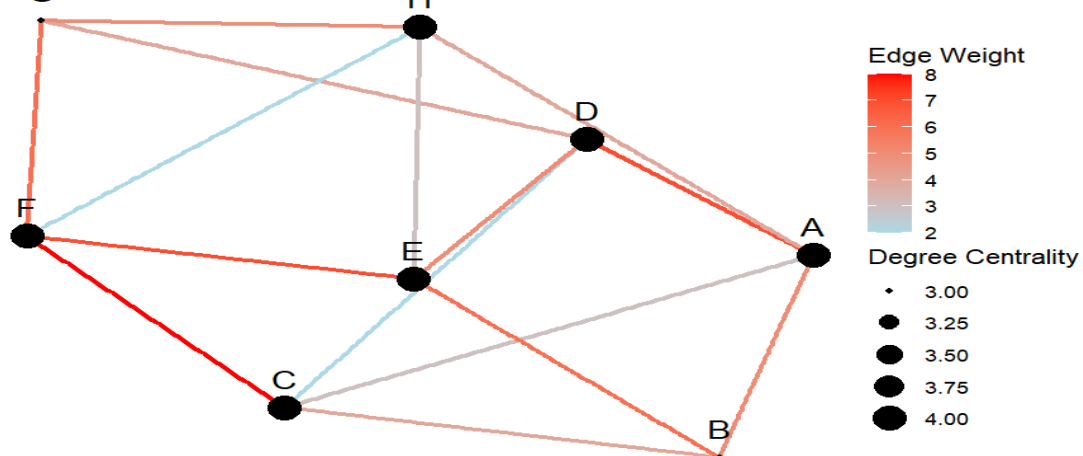
The plot shows a positive relationship between X and Y. A hexbin plot is preferred because 10,000 observations would cause significant overplotting in a normal scatterplot. Hexagonal bins summarize dense regions clearly.

Why Hexbin is Better Here

Standard Scatterplot	Hexbin Plot
Many points overlap	Groups points into hexagons
Difficult to identify dense regions	Clearly shows density
Overplotting occurs	Reduces overplotting
Less useful for very large datasets	Suitable for large datasets

Question 5(B): Network Graph

Network Graph with Degree Centrality



CODE:

```
# Q5B - Network Graph

library(igraph)

library(gggraph)

library(ggplot2)

# Create edge-list dataset

edges <- data.frame(

  from = c("A","A","A","B","B","C","C","D",

           "D","E","E","F","F","G","H"),

  to = c("B","C","D","C","E","D","F","E",

         "G","F","H","G","H","H","A"),

  weight = c(5,3,7,4,6,2,8,5,4,7,3,6,2,5,4))

# Display edge data

print(edges)

# Create undirected graph

g <- graph_from_data_frame(edges, directed = FALSE)

# Calculate degree centrality

degree_values <- degree(g)

# Add degree centrality to nodes

V(g)$degree <- degree_values

# Find node with highest centrality

highest_node <- names(which.max(degree_values))

cat("Degree Centrality:\n")

print(degree_values)
```

```

cat("\nNode with Highest Centrality:",
    highest_node, "\n")

# Create network graph

ggraph(g, layout = "fr") +

geom_edge_link(
    aes(color = weight),
    width = 1 ) +

geom_node_point(
    aes(size = degree) ) +

geom_node_text(
    aes(label = name),
    vjust = -1,
    size = 5) +

scale_edge_color_gradient(
    low = "lightblue",
    high = "red" ) +

labs(
    title = "Network Graph with Degree Centrality",
    edge_color = "Edge Weight",
    size = "Degree Centrality") +

theme_void()

```

Interpretation

The node printed by:

`highest_node`

is the node with the **highest degree centrality**. This means that the node has the largest number of direct connections with other nodes in the network. Therefore, it can be considered an important **hub node**, as it may play a major role in connecting different parts of the network and facilitating interaction or information flow.

Important: Since your actual `network_edges.csv` file is not included here, the exact highest-centrality node cannot be stated without seeing the edge list. Run the above code, and R will automatically print the correct node.